



UNIVERSITY OF PISA

DEPARTMENT OF COMPUTER SCIENCE

Data Mining Report

Group 7

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Chapter 1

Data Understanding and Preparation

The first task of the project consists in analyzing the datasets given and preparing them for the future tasks. First, a deep analysis of the structure of each dataset was performed, discovering inconsistencies and erroneous values. Second, some cleaning steps and preparation steps were performed to enable the subsequent tasks, such as clustering and classification.

1.1 Data Understanding

Of the three datasets, the one that raised the most problem is *ncidents*, because it was the only one containing NaNs, wrong types of data and values out of range for some attributes. Most of the understanding and the cleaning/preparation was performed on this dataset for this exact reason.

1.1.1 NaN and Type analysis

Most of the columns were poisoned with NaN values, and the worst ones were *min_age_participants*, *avg_age_participants* and *max_age_participants*, with more than 30% of the values as NaN.

The *(min/max/avg)_age_participants* and *n_participants_(child/teen/adult)* were considered object while in reality they should have had numeric dtype, suggesting a mixing of types. The features regarding the age of the participants had roughly 3.5% of non-numeric and nonNaN values that were not convertible to numeric.

1.1.2 Columns Filtering

For the Poverty and State District House datasets all the columns were kept while for *incidents* only the following were not dropped:

- date

- state
- city_or_county
- latitude
- longitude
- congressional_district
- n_participants_child
- n_participants_teen
- n_participants_adult
- n_males
- n_females
- n_killed
- n_injured
- n_arrested
- n_unharmed
- n_participants
- incident_characteristics1

1.1.3 Values and Range Check

Before the preparation part of the Task, it was necessary to check whether some of the features were correct.

All the dates strings followed the format (YYYY-MM-DD), but some dates were in the future, so out of the possible range.

No errors were found in the names of the States. Washington DC, even though is not a state, was included in the dataset, but it was removed due to the low number of incidents.

Latitude and Longitude values were also in the correct range respectively $(-90, 90)$ and $(-180, 180)$.

Regarding the participant age a couple of erroneous values were found (311 and 209 which are not valid ages) and also some values of `(min|avg|max)_age_participants` were outside ranges.

1.1.4 Correlations

The last interesting thing before starting to work on the datasets is to show the correlation between all the features, by using the Correlation Matrix (shown in figure 1.1).

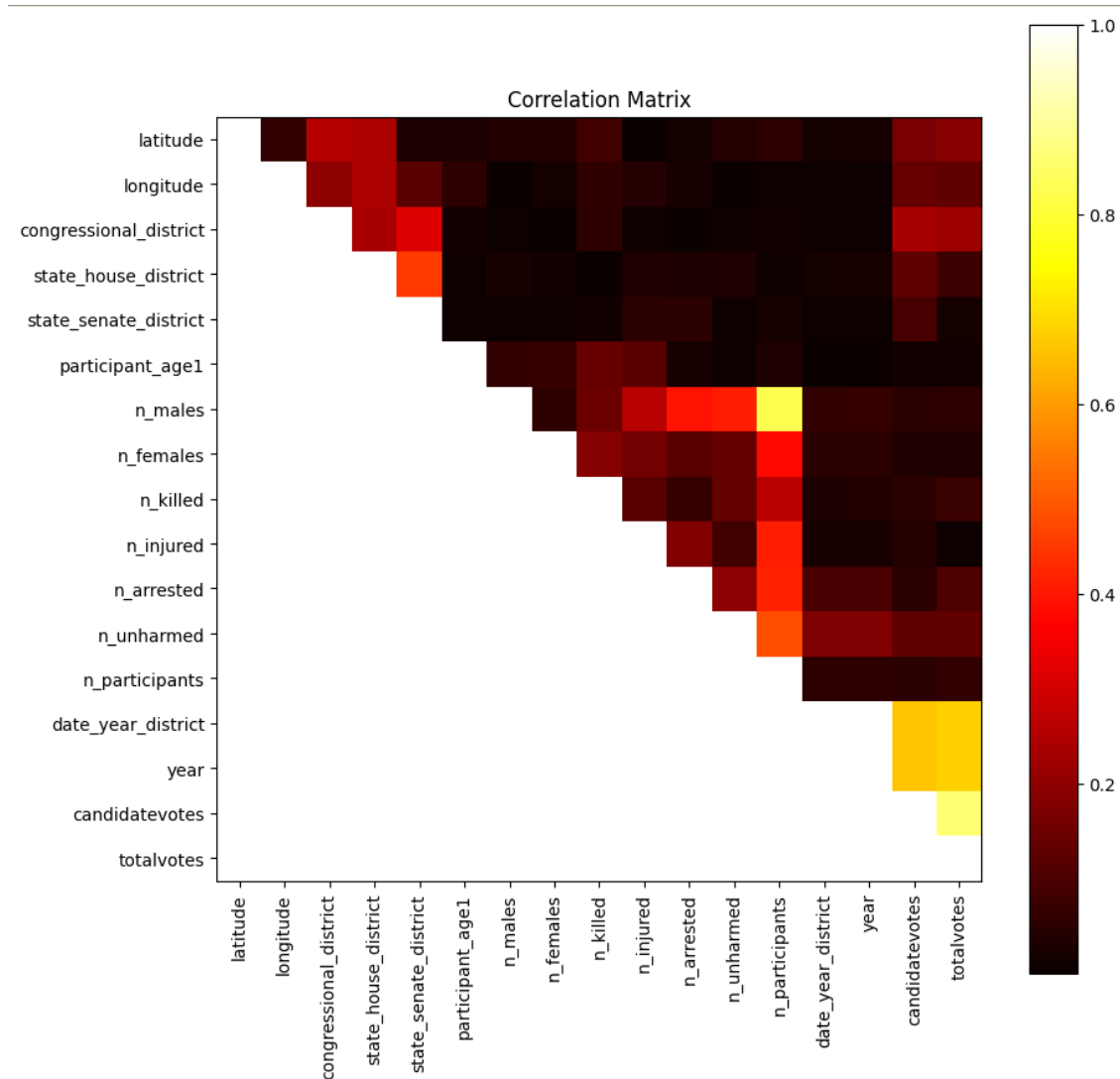


Figure 1.1: Correlation Matrix

As shown, there is not an high correlation between any of the features, except the number of males participants with the total number of participants in a given incident.

1.2 Data Preparation

1.2.1 Data Cleaning

The data cleaning process can be summarized in 9 main steps:

1. Remove the rows with 0 participants.
2. Merge incidents, poverty and state district house data sets.
3. Set to NaN all the cells that are supposed to be numeric values and they are not.
4. Set out-of-sensible range values to NaN.
5. Infer the congressional district when missing as the most common one given the state and the city_or_county values (i.e. sample the congressional_district value from $P(\text{congressional_district} \mid \text{state}, \text{city_or_county})$).
6. Given the groups
 - n_participants,
 - n_males, n_females,
 - n_participants_child, n_participants_teen, n_participants_adult,
 - n_killed, n_injured, n_arrested, n_unharmed,

find and solve inconsistencies between their sums, keeping the original proportions and assuming the most common sum as the correct one (or NaN if all differents).

Example:

- n_participants = 10,
 - n_participants_child + n_participants_teen + n_participants_adult = 10,
 - n_males = 2, n_females = 6, then the most probable values (according to our assumptions) are groups that sum to 10 so: n_males := 3, n_females := 7.
7. For each row, identify columns that have missing values, filter the DataFrame to include only points where the non-missing values of the row match the corresponding ones in the point and sample a row from the filtered DataFrame, and updates the input row with the sampled values for the missing columns (i.e. sample the new row from $P(\text{missing values} \mid \text{non missing values})$).
 8. Drop all irreparable rows.
 9. Drop duplicates.

Chapter 2

Clustering

2.1 Common Setting

The data preparation common to all different clustering algorithms consists of three main steps: categorical attributes dropping, outlier removal and scaling. The removal of outliers has been performed by applying the IQR method, i.e. cutting all the points that have at least one attribute whose value falls out of the IQR of the column; the threshold value used for the computation of the IQR is the default one of 1.5.

Then the remaining points have been scaled with the Min-Max scaling method for two main reasons:

- Empirically, it leads to better clustering results with respect to the Standard scaling method, maybe due to skewed (asymmetric) or not approximately Gaussian distributions of some defined indicators and to features with very different scales and variances. Using the Min-Max scaling guarantees that all the attributes values fall into the same range $[0, 1]$ ensuring a consistent scale across features.
- The fact that the scaled values are in a specific range $[0, 1]$ makes them easily interpretable.

All the explored clustering approaches, respectively K-means, density-based (DBSCAN) and hierarchical have been applied to different data spaces:

- the data space defined only by the original attributes (without the defined indicators),
- the data space defined by both original attributes and indicators,
- the data space defined by only the defined indicators (without the original attributes),
- the data space obtained applying dimensionality reduction (attributes cutting based on variance and correlation) to each of the preceding spaces.

In the experiments involving indicators, the outlier removal has been performed before the augmentation with the indicators (i.e. only on the original features), in order to compare the performances on the same set of points represented on different spaces.

The difference in results between the original data space and the augmented data space measures the impact of the defined indicators on the clustering, while the difference in results between the indicators data space and the data space containing original attributes measures how much of the useful original information the indicators contain/preserve.

The dimensionality reduction based on variance and correlation consists in two steps:

1. cut all the features that have a very low variance in the entire dataset;
2. cut the features that have a high correlation with other features.

The reduction is applied on the scaled data, using a variance threshold of 0.01 and a correlation threshold of 0.25.

Each clustering scenario is evaluated by comparing the resulting silhouette score with the silhouette score of the same clustering algorithm applied on a randomly generated dataset in the same data space.

A random dataset containing the same number of points of the true dataset without outliers is generated in the original feature space, respecting the value range and the type of each attribute. Then the augmented dataset and the indicators dataset are computed from this random data.

The clustering algorithms have been performed using euclidean distance as a metric. Finally an interpretation of the resulting clusters is given looking at the distribution of attributes inside the clusters.

2.2 K-means

Note: The same clustering procedure is applied to each data space.

2.2.1 K Selection and Evaluation

One of the choices to perform for applying K-means clustering is the choice of the number of clusters k . In the experiments the tried values are the ones in the interval $[2, 20]$. For each k a K-means clustering is performed and a silhouette score is computed; while the clustering is performed on the entire set of points, the silhouette score is computed for samples of points, since it requires too much time. After a first computation of the score, the best k value (the one with the highest score) and the near values are selected, assuming a sort of locality property, and another silhouette score computation is performed on them, increasing the sample sizes.

Finally only the best k is selected and a final silhouette score computation is performed on it, further increasing the sample sizes, in such a way to have a more

accurate estimate of the silhouette on the whole set of points.

In Table 2.1 are reported the number of samples and sample sizes of each silhouette score computation.

Number of clusterings	Number of samples	Sample sizes
=19	5	1000
< 5	3	5000
= 1	3	10000

Table 2.1: Samples details for silhouette score computation

2.2.2 Results

In Figures 2.1 and 2.2 are summarized the clustering results.

The clustering applied to the data spaces reduced using variance and correlation informations gives better results both in terms of absolute silhouette score and in terms of score difference from the random case.

Moreover the reduced scenarios show lower variances in the silhouette score computations than the full data space score computations.

Data space	k	Silhouette mean	Silhouette variance	Silhouette difference from random data clustering
Original	2	0.341907	2.716524e-06	0.262753
Augmented	4	0.273376	8.380057e-06	0.176836
Indicators	3	0.292416	1.092029e-06	0.170739
Original Reduced	2	0.415102	7.220957e-07	0.335372
Augmented Reduced	4	0.385371	5.471591e-07	0.324141
Indicators Reduced	2	0.453778	3.048582e-08	0.301485

Figure 2.1: Table of k-means results

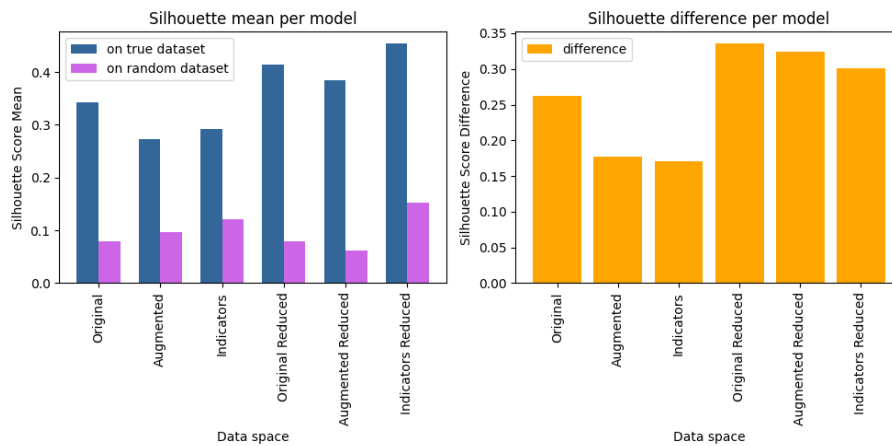


Figure 2.2: K-means scores

2.2.3 Analysis

Looking at the centroids, it seems attributes regarding male, injured and adult participants are distinguishing aspects that differ the centroids of the highest score clusterings. These aspects can be observed also looking at the male, injured and adult attributes distributions in the clusters.

For the analysis of categorical distributions among clusters, we consider the clustering on the reduced indicator data space (Figure 2.3), which is the best in terms of score and third best in terms of difference with the random case.

Best clustering: Reduced indicators data space

Cluster 0: 59466

Cluster 1: 35725

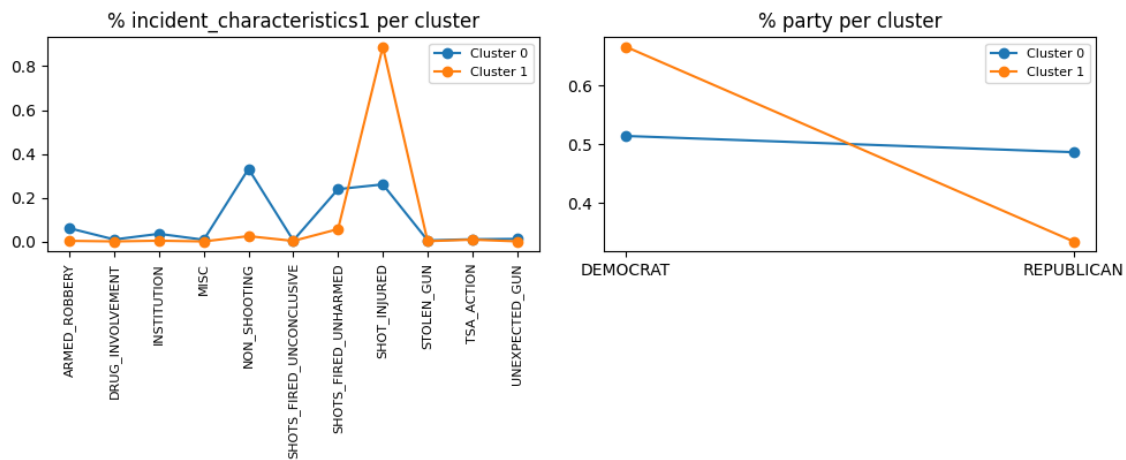


Figure 2.3: Categorical distributions per cluster (indicator reduced data space): The most consistent differences in the per cluster categorical distributions are the ones of the incident characteristics and party attributes. Cluster 0 has higher percentages of ‘armed robbery’, ‘non shooting’ and ‘shot fired unharmed’ characteristics, while cluster 1 has a higher value for the shot injured characteristic. Finally the party distributions show a relevant percentage of democratic party over the republican party in cluster 1, while in cluster 2 the distribution is more uniform and balanced (closer to 0.5).

2.3 DBSCAN

2.3.1 Parameters Selection

The DBSCAN algorithm requires to set two parameters: ϵ (the neighborhood radius) and n (the minimum number of neighbors for a point to be considered a central point). As for the k parameter of k -means, a sort of locality property on the n parameter is assumed. Based on that, a greedy grid search is performed on n ;

each state (n) is evaluated using the best silhouette score obtained varying the ϵ parameter. Fixed n , the ϵ that maximize the score is selected.

2.3.2 Results

Model	Silhouette	N_clusters	Cluster_sizes	Clusters
Original Model	0.512991	3	[2088, 1758, 2180]	[0, 1, 2]
Augmented Model	0.363923	3	[1755, 2198, 2073]	[-1, 0, 1]
Indicators Model	0.398063	2	[3, 6023]	[-1, 0]
Original Reduced Model	0.670266	2	[2088, 3938]	[0, 1]
Augmented Reduced Model	0.449460	2	[3, 6023]	[-1, 0]
Indicators Reduced Model	0.258194	2	[830, 5196]	[-1, 0]

Figure 2.4: DBSCAN results

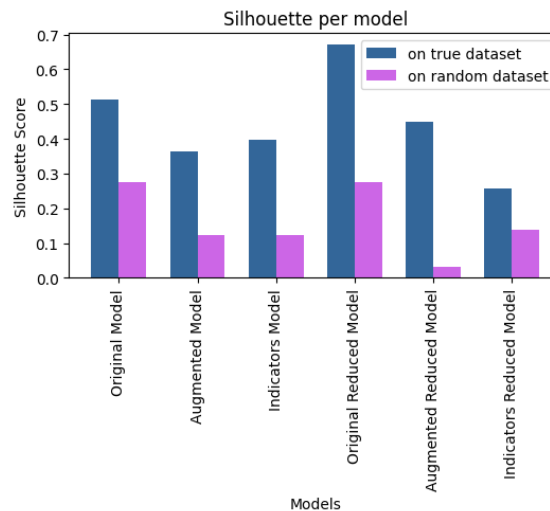


Figure 2.5: DBSCAN results

2.3.3 Analysis

Also if DBSCAN does not perform a centroid based clustering, the mean of each founded cluster has been computed and then a filtering has been performed on the features, selecting only the attributes with a low variance (< 0.1) inside each cluster and such that the highest absolute difference between the centroids attribute values is above a certain threshold (0.5). The variance constraint guarantees the representative power of the filtered features, while the distance constraint select features with enough discriminative power. For the best clustering (the one on the original reduced data space), the only survived feature is the number of killed, meaning that it is the most (and only enough) representative and discriminative attribute.

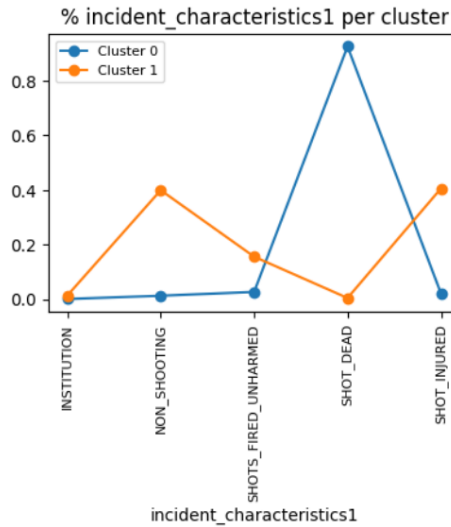


Figure 2.6: The most relevant differences in the distributions of the categorical attributes regards the incident characteristics. Cluster 0 has an higher percentage for 'shot dead', while cluster 1 has higher values for the 'non shooting', 'shot fired unharmed' and 'shot injured' characteristics.

2.4 Hierarchical Clustering

The explored algorithm for hierarchical clustering is the agglomerative one. The algorithm has been tested with all the available methods ('single', 'complete', 'ward', 'average', 'weighted', 'centroid', 'median').

The best cut (i.e. the cut that gives the highest silhouette score) in the resulting dendrogram of each method has been searched in the highest 10 layers (excluding the single-cluster layer).

2.4.1 Results

dataset	best method	silhouette mean	silhouette variance	silhouette max	silhouette min	best height	height mean	height variance	height max	height min	best k	k mean	k variance	k max	k min
Original	single	0.512991	0.000000	0.512991	0.512991	0.502682	4.228152	56.665282	22.630433	0.502682	3	3.000000	0.000000	3	3
Augmented	ward	0.359197	0.000183	0.371067	0.329292	48.477395	9.167098	258.037278	48.477395	1.622095	3	4.428571	1.102041	6	3
Indicators	single	0.384085	0.000745	0.409695	0.328230	2.295832	6.544439	84.262738	28.996041	2.097963	2	2.571429	1.102041	5	2
Original Reduced	single	0.670266	0.000000	0.670266	0.670266	0.196303	3.175409	38.437276	18.342744	0.196303	2	2.000000	0.000000	2	2
Augmented Reduced	single	0.552343	0.000003	0.553358	0.549804	0.724536	3.915721	47.014500	20.695772	0.724536	2	2.285714	0.204082	3	2
Indicators Reduced	ward	0.436765	0.000378	0.454514	0.390702	17.613063	3.460279	33.553467	17.613063	0.754810	5	5.714286	5.632653	9	2

Figure 2.7: For each data space, the silhouette, height and k statistics computed among the tested methods and the model that leads to the best score

2.4.2 Analysis

Best clustering: Reduced original data space

Cluster 0: 3938

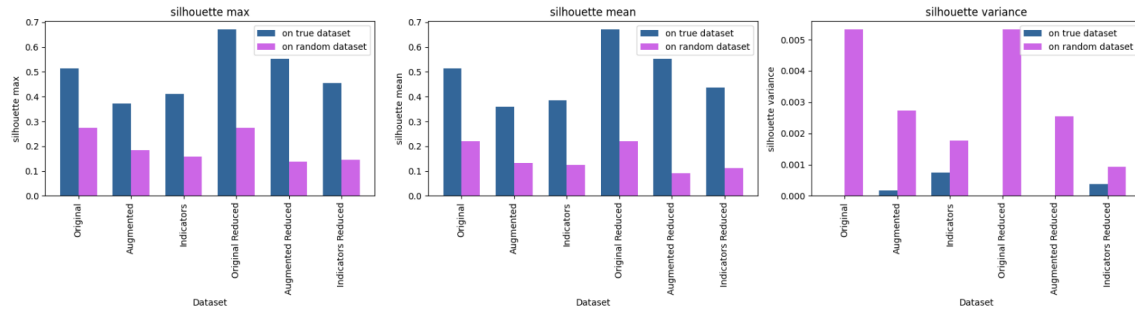


Figure 2.8: Results statistics compared with the random case

Cluster 1: 2088

The same centroids analysis applied for DBSCAN has been performed. For the best clustering (the one on the original reduced data space), the only survived feature is, once again, the number of killed.

The best resulting clustering shows very similar characteristics of the best clustering resulting from DBSCAN, in terms of clusters number, clusters sizes, silhouette score and in terms of categorical attributes distributions among clusters.

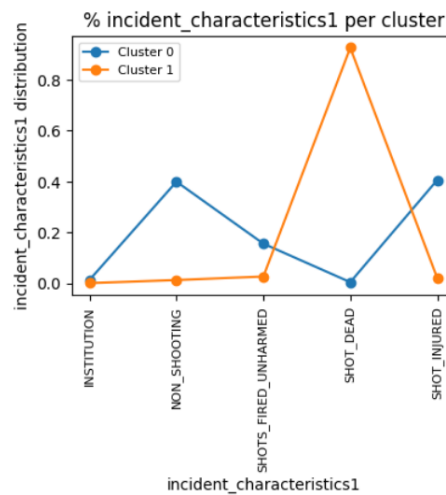


Figure 2.9: As for DBSCAN, the most relevant differences in the distributions of the categorical attributes regards the incident characteristics. Cluster 1 has an highest percentage for 'shot dead', while cluster 0 has higher values for the 'non shooting', 'shot fired unharmed' and 'shot injured' characteristics.

Chapter 3

Classification

In this task, our goal was to create a classifier which, given an incident, would predict the presence of a fatality.

Two different classifiers have been tested: a Decision Tree, for its interpretable nature, and a Neural Network, for its ability to capture complex relationships between the features.

3.1 Methods

Since the target variable comes from the dataset in the form of `n_killed` which is the number of fatalities, it was necessary to transform it into a binary variable (`isKilled`) and to drop the original one and all the features directly deriving from it.

3.1.1 Feature Selection

Some early tests showed nearly perfect results from the classifiers. Analyzing the Decision Tree it was easy to see that the classifier was obtaining the value of `n_killed` from `n_participants - n_injured - n_unharmed - n_arrested`.

To avoid this problem, `n_participants` and all the features that would allow to obtain it were dropped.

Later tests showed that the classifier was still *cheating* when `n_injured == n_unharmed == n_arrested == 0`; in this case, given that `n_participants ≥ 1`, the value of `n_killed` would be ≥ 1 . To avoid artificially inflating the scores, the datapoints where this condition was met were dropped.

3.1.2 Scaling

For the Decision Tree classifier, no scaling was necessary, so it was not performed to keep the interpretability of the model.

For the Neural Network, the dataset was scaled using the `StandardScaler` from SKLearn.

3.1.3 Training-Validation-Test split

The dataset was split into three parts: training (70%), validation (15%) and test (15%). The training set was used to train the classifier, the validation set was used to evaluate the difference in performance between different classifiers and to tune the hyperparameters, and the test set was used to evaluate the final performance of the classifiers.

3.1.4 Balancing

The dataset is unbalanced, with around 75% of incidents having no fatalities. This could lead to a classifier that always predicts no fatalities. To avoid this problem the dataset was balanced using the [SMOTE](#)[?] algorithm on the training set, and with upsampling with replacement on the validation and test sets (to avoid testing on synthetic data).

3.2 Decision Tree

SKLearn's Decision Tree classifier was used. To try to avoid overfitting, a Grid Search was performed.

Despite this, the best classifier obtained had an F1 score of 0.93 on the training set, while on the validation set it had an F1 score of 0.72.

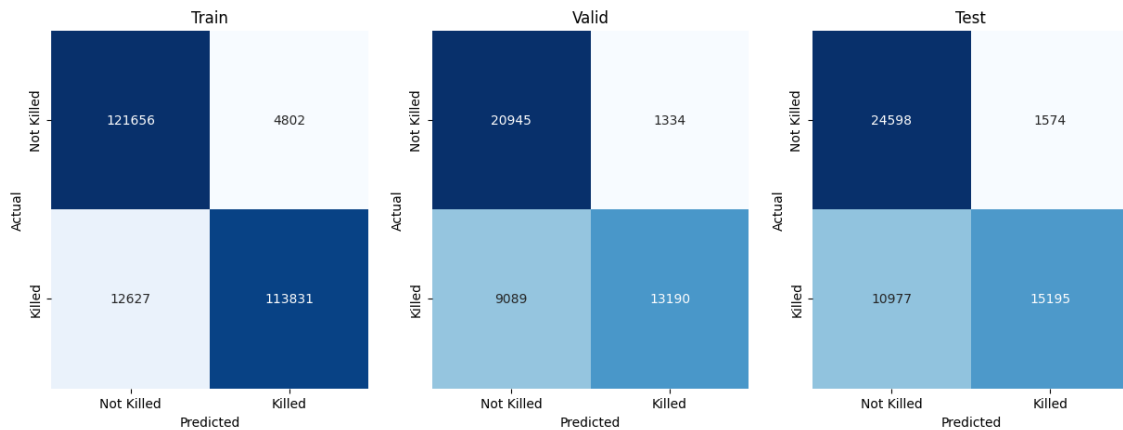


Figure 3.1: Decision Tree confusion matrix of the training, validation and test sets.

On Figure 3.1 the confusion matrices of the training, validation and test sets can be analyzed. While still having positive results, the classifier is clearly overfitting.

The final test F1 score was 0.71. Figure 3.2 shows the first three levels of the Decision Tree.

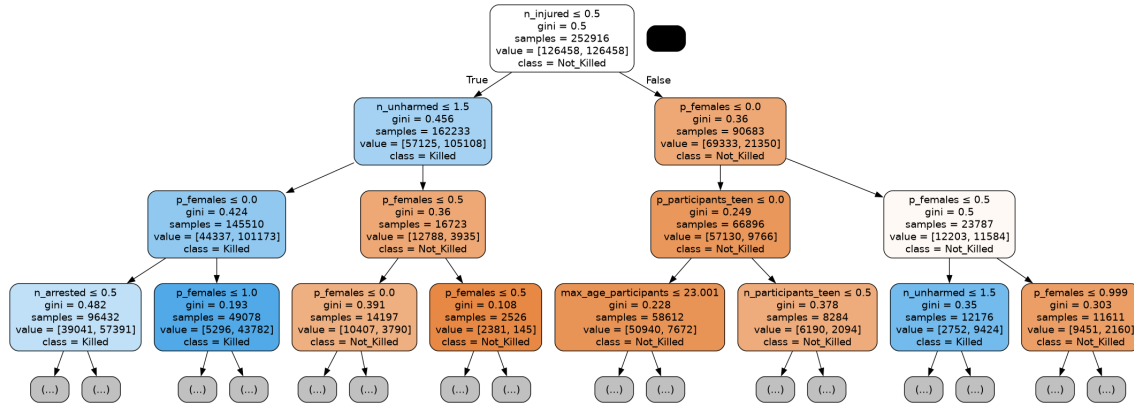


Figure 3.2: Decision Tree.

3.3 Neural Network

A Neural Network was trained hoping that it would find hidden relations in the data. The architecture used was the following:

- Input layer with 23 neurons (one for each feature)
- 4 Hidden layers with [32, 16, 8, 4] neurons and Sigmoid activation function, followed by a 10% dropout rate
- Output layer with 2 neurons and Softmax activation function

The model was trained with an Early Stopping callback, which stopped the train- ing if the validation score did not improve for 10 validation steps.

The Early Stopping kicked in at the 13000th epoch, with a training F1 score of 0.82 and a validation F1 score of 0.78.

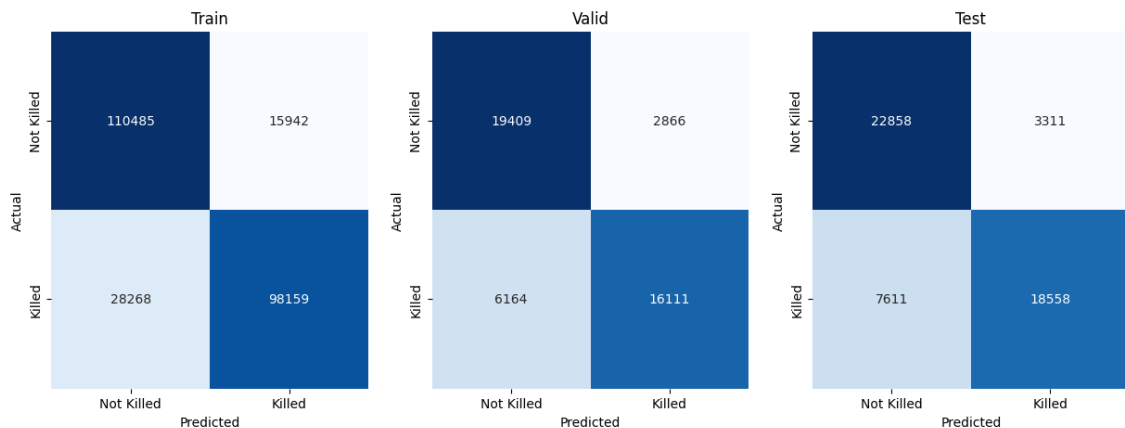


Figure 3.3: Neural Network confusion matrix of the training, validation and test sets.

Figure 3.3 shows the confusion matrices of the training, validation and test sets. The final test F1 score was 0.77.

3.4 Comparison

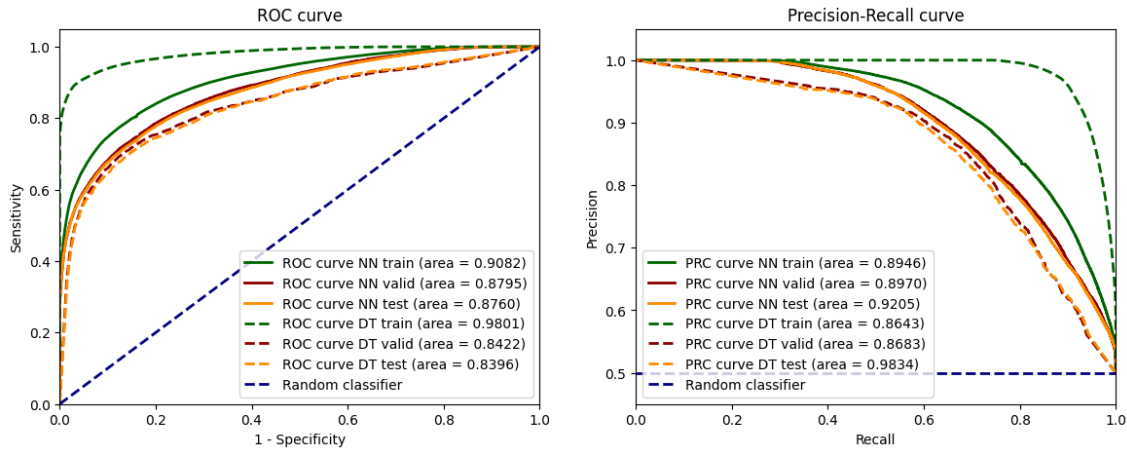


Figure 3.4: ROC and Precision-Recall curves of the Decision Tree and Neural Network compared.

Figure 3.4 shows the ROC and Precision-Recall curves of the Decision Tree and Neural Network compared.

The Neural Network has a slight advantage over the Decision Tree, with higher scores on the test set, but not significant enough to mark it as better suited for the task.

It is important to note that due to hardware limitations, the search for the best hyperparameters was not exhaustive, and it is possible that better results could be obtained with a more thorough search.

It is important to note that due to hardware limitations, the search for the best hyperparameters was not exhaustive, and it is possible that better results could be obtained with a more thorough search.

Chapter 4

Time Series Analysis

The goal of this task is to generate a number of timeseries starting from a score defined for each city and for each week. Then clustering algorithms are applied on the timeseries to group them in clusters. It is also performed a Motif and Anomalies analysis on some of the timeseries to search for recurring pattern on each of those. Finally, after defining a new score for each city, a Shapelet discovery was performed to classify all the timeseries based on the defined variable *isKilled*.

4.1 Dataset Preparation

To start the analysis it was necessary to prepare the dataset. The starting dataset, called *final_clean_3*, was first filtered by only retaining the incidents between 2014 and 2017, and then grouped by week number and city. Finally, the dataframe *clean_city_df* was created, in which only the cities with a number of incidents greater than the 15% of incidents in the 209 weeks total.

4.2 City Score

For the first analysis on Motif and Anomalies, it was necessary to define a score for each city. The main idea behind this score is to measure the severity of the incidents for that week in each city. This was done with a weighted sum of various indicators, with the weights chosen arbitrarily in order to allow to change them based on the importance that each indicator can have. The score is defined as follows:

$$\begin{aligned}
\text{Score} = & n_killed_week \\
& + n_injured_week * 0.8 \\
& + n_arrested_week * 0.5 \\
& + n_child_week \\
& + n_teen_week * 0.8 \\
& + n_adult_week * 0.5
\end{aligned}$$

n_killed_week = number of killed people in that week

$n_injured_week$ = number of injured people in that week

$n_arrested_week$ = number of arrested people in that week

n_child_week = number of child participants in that week

n_teen_week = number of teen participants in that week

n_adult_week = number of adult participants in that week

A higher weight was given to the number of killed and injured people, and a lower weight to the number of arrested people. A higher weight was also given to the number of children and teenagers involved in the incident, and a lower weight to the number of adults involved.

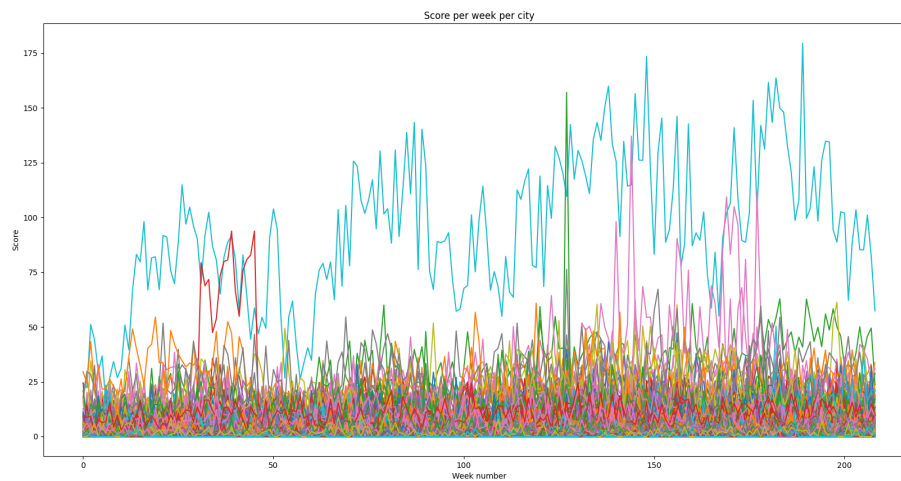


Figure 4.1: Time Series generated with the score of all the cities

4.2.1 Time Series Scaling

Most of the analysis for the time series were conducted not only on the original ones, but also on a scaled version of those. The two scaler used were the Standard scaler and the MinMax scaler. Figure 4.2 and figure 4.3 show the plot of the two time series.

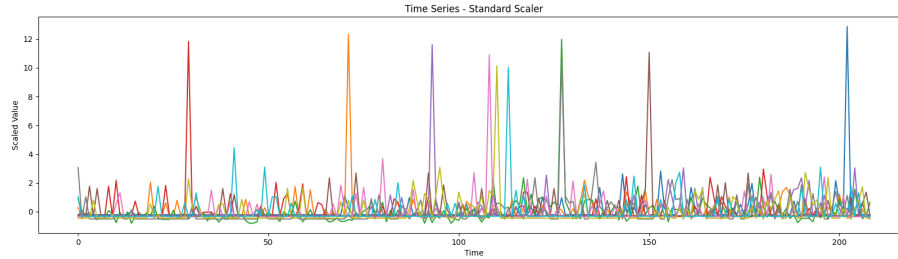


Figure 4.2: Time Series scaled with the Standard Scaled

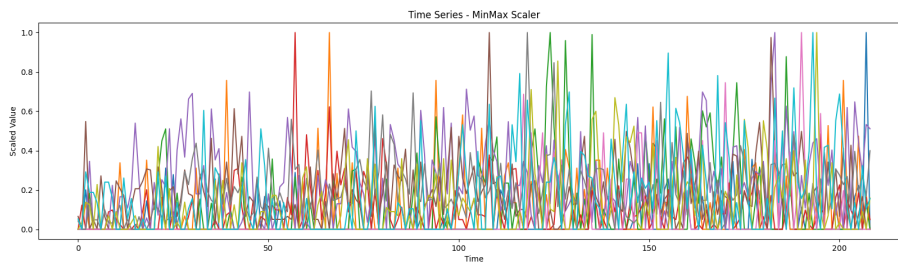


Figure 4.3: Time Series scaled with the MinMax Scaled

4.3 Time Series Clustering

Due to the fact that most of the Clustering algorithms were already tested in depth in Task_2, only K-means was used for this analysis. The clustering analysis was also divided in two different ones: One conducted on the real time series and one on the feature extracted from them.

4.3.1 Clustering on the real time series

For all the applications of K-means only the Dynamic Time Warping (DTW) distance was used, mainly because it is better suited to capture time series with varying pattern, time shifts or speed, all things that the euclidean distance could miss.

To find the best number of cluster on which divide the time series the same method used in Task_2 was used, i.e. the Silhouette score.

The results in table 4.3.1 shows that the clustering on the non-scaled time series reached a way higher score than the scaled ones.

n_clusters	sc_original	sc_minmax	sc_standard
0	0.812281	0.190895	0.001042
1	0.781509	0.023115	-0.000491
2	0.625345	-0.010280	-0.001191
3	0.458336	-0.014969	-0.001957
4	0.458559	-0.034263	-0.001939
5	0.461002	-0.037447	-0.003702
6	0.424799	-0.018583	-0.004466
7	0.413574	-0.047496	-0.005049

Table 4.1: Silhouette Score for the different Scaling

4.3.2 Clustering on the feature of the time series

Another technique often used when dealing with time series analysis is to reduce them to a vector containing some significant features, such as mean, variance, kurtosis ecc... and then apply the clustering algorithms to those points.

This analysis was also conducted on the original time series and the scaled ones.

As seen in table 4.2 the silhouette score was overall higher than the one on the real time series. A significant result is the score on the non scaled features that reached a .98 with 2 clusters

n_clusters	sc_original	sc_standard	sc_minmax
2	0.984422	0.620615	0.674923
3	0.936071	0.587219	0.573063
4	0.872255	0.469417	0.513506
5	0.825605	0.426069	0.501098
6	0.765932	0.390568	0.475368
7	0.765437	0.373845	0.395098
8	0.689629	0.355427	0.380166
9	0.690026	0.357124	0.378075

Table 4.2: Silhouette Score on the features for the different Scaling

4.3.3 Comparison of the clustering with a random walk

One of the strategy to see how well the clustering algorithm works, is to compare the score of the clusters found with the ones of a random walk.

A random set of time series was generated with the same number of points of the original set, and the min and max values of those time series was also the same as the one found with the score.

Table 4.3 shows the silhouette score only for the non-scaled random time series, but it is clear that the clustering on those time series performed really poorly, with the best score reaching .001.

n_clusters	sc_random
2	0.001042
3	-0.000491
4	-0.001191
5	-0.001957
6	-0.001939
7	-0.003702
8	-0.004466
9	-0.005049

Table 4.3: Silhouette Score on the random time series

n_clusters	sc_random_F
2	0.547966
3	0.510407
4	0.511698
5	0.483421
6	0.481377
7	0.485476
8	0.478391
9	0.480326

Table 4.4: Silhouette Score on the random time series features

Regarding clustering on the feature of the random walks, the clustering performed better (Table 4.3), but it was still way lower than the score obtained on the original time series.

4.4 Motif and Anomaly Detection

To conduct an analysis on Motif and Anomalies, it was necessary to choose some time series to study. The ones chosen for this task were the cluster centers found with K-means in the previous section.

Although the significance of those time series could be discussed, the idea behind it is that no other time series had a bigger significance than those.

The two centroids are kinda representative of the time series in each cluster, and that is why were chosen

For the sake of testing, two of the time series with the highest score and the one with the lowest overall were also analyzed (but not in details).

Another possible analysis could be conducted on a small sample from each of those clusters, but still this analysis is not that much more significant.

The interpretability of the Motifs found (figure 4.4) is not particularly clear, but it is shown that, contrary to the two centroids, on the random walk only one big motif was found, meaning that no real repetitiveness could be found.

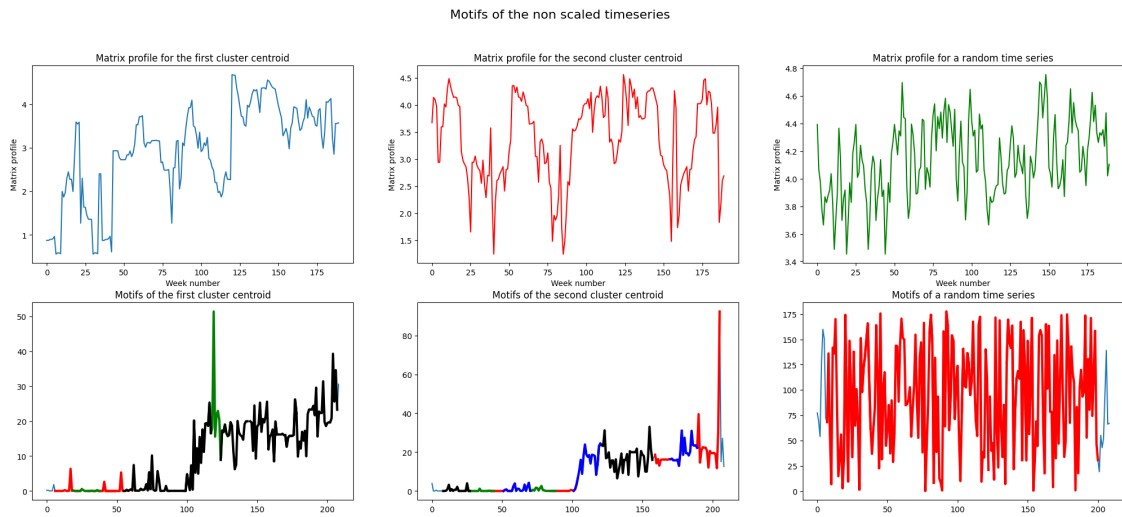


Figure 4.4: Motifs of the two non-scaled centroids and the random walk

Not much more could be extrapolated on the three cities, with only Chicago showing some sort of annual seasonality on the motifs. (figure 4.5)

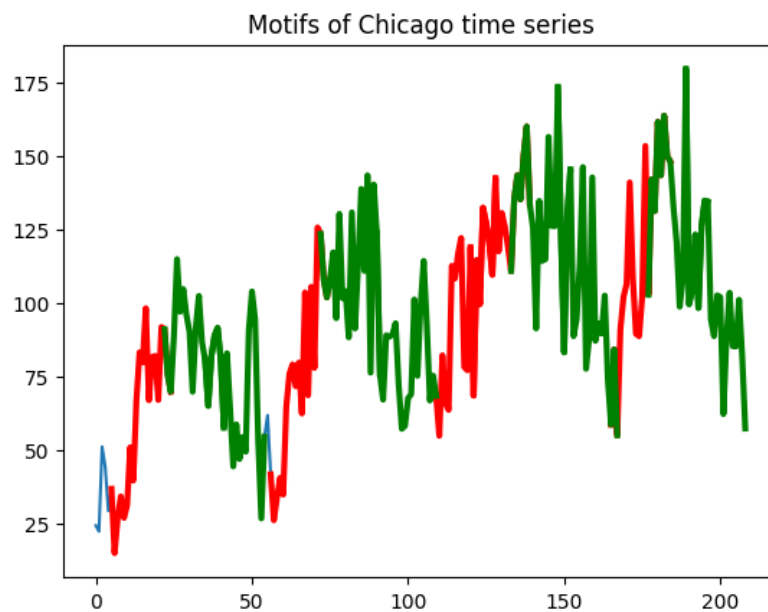


Figure 4.5: Motifs of Chicago

As far as Anomaly detection, we conducted it on the non scaled time series, the three cities and the random walk, and we were able to find a couple of anomalies, but these anomalies cannot also really be interpreted in a clear way.

4.5 Shapelets

For the analysis on the shapelets of the time series, it was required a score that was not directly correlated with the variable *isKilled*. The new score for each city is defined as follows:

$$New_Score = n_incidents_week + n_participants_week$$

$n_incidents_week$ = number of incidents in that week

$n_participants_week$ = number of participants of all the incidents in that week

The idea behind this score is that there might be an indirect correlation between the number of incidents and the number of participants and the class *isKilled*.

Figure 4.6 shows the time series generated in this way.

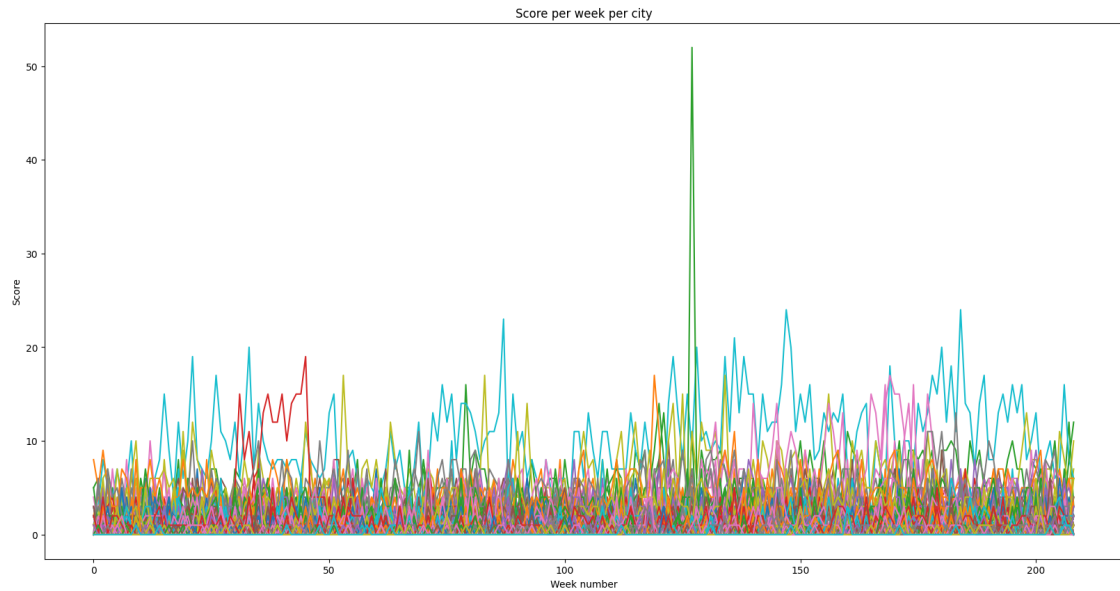


Figure 4.6: Time series generated with the new score

The threshold after which a time series is classified as a positive value for *isKilled* was determined in a way to have almost the same number of cities in both of the classes.

For the shapelets extraction, only a small number of time series were sampled (4 to be precise), due to the high computational cost of the algorithm.

Unexpectedly, even with such a small number of series on which the model was trained, the training ROC curve was optimal ($auc = 1$) and for all the other time series not used for training the ROC curve was still good.

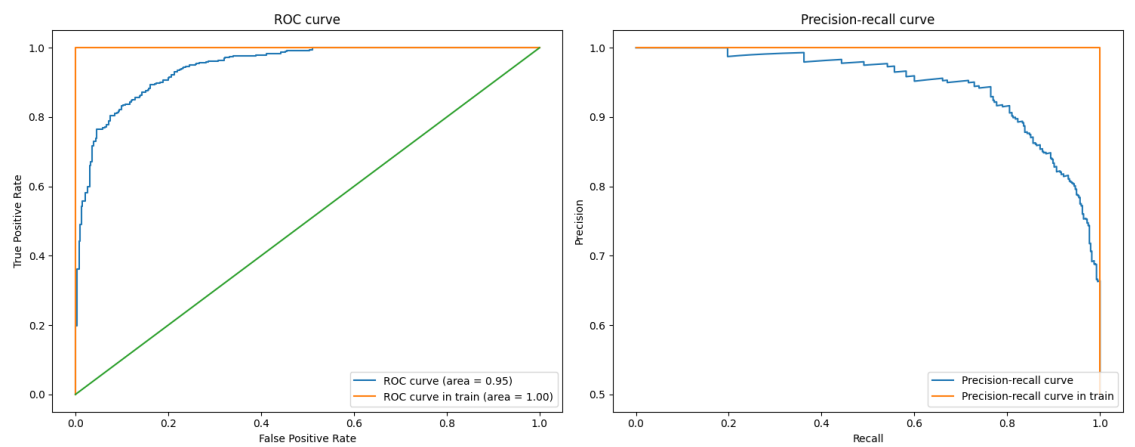


Figure 4.7: ROC curve for the train and test set

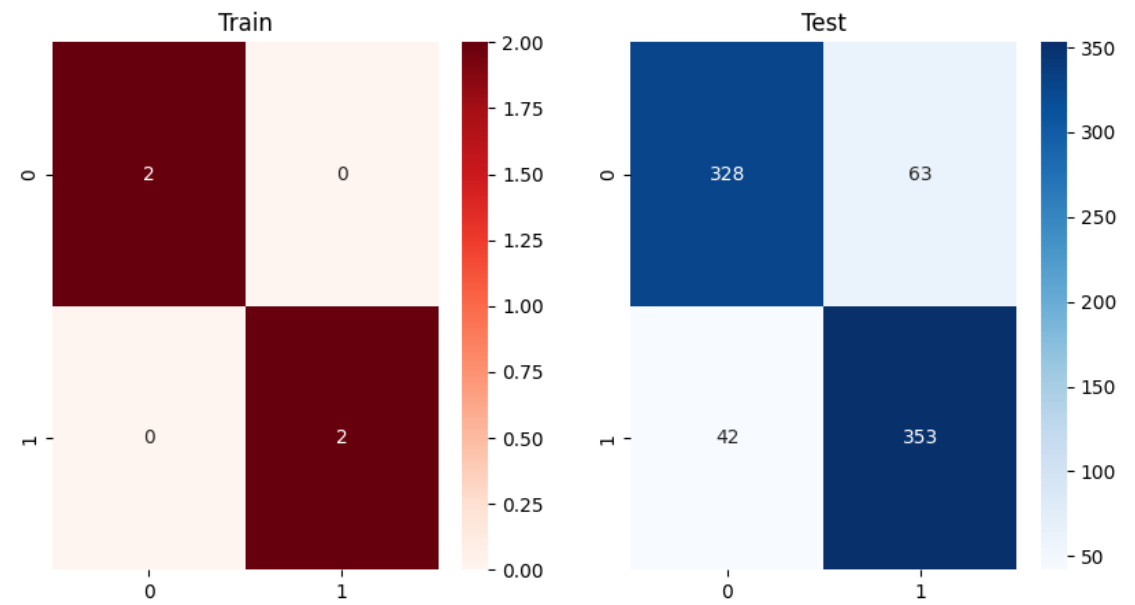


Figure 4.8: Confusion Matrix for the train and test set